

FIT2109 Computer Science Workshop

Week 0: Welcome & Unit Overview

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Acknowledgement

I wish to acknowledge the people of the Kulin Nations, on whose land we are gathered today. I pay my respects to their Elders, past and present.

Welcome to FIT2109

What is this unit?

FIT2109 Computer Science Workshop teaches the tools and practices professional software developers use every day — the things that make the difference between writing code that works and shipping software that lasts.

One sentence version

“Twelve weeks of things your other units assume you already know.”

By the end of semester

- Comfortable at the command line.
- Git workflows that scale to teams.
- Containerised, reproducible environments.
- Systematic debugging and profiling.
- Automation, scripting, and agentic tools.

Teaching team

Unit coordinators / Lecturers

Yongqiang Tian Unit Coordinator / CE
Weeks: 1, 2, 6, 8, 9, 12

Zahra Lecturer
Weeks: 3, 4, 5, 7, 10, 11

Teaching associates

Your applied-class TA is your first contact for hands-on setup and worksheet questions. Session allocations and consultation times are listed on Moodle.

How to get help

- ① Try it yourself first — that is the point of the unit.
- ② Ed Discussion board (fastest; others benefit too).
- ③ Applied class drop-in with your TA.
- ④ Lecturer consultation times, posted on Moodle.

Response time

Ed posts answered within 48 h on weekdays.
Email is for personal matters only — use Ed for all content questions.

Topics we will cover

Core tool fluency

- Shell and Bash fundamentals
- Shell tools, regex, scripting, and SSH
- Editor and modern IDE workflow
- Git foundations, collaboration, and history
- Command-line environment

Building reliable software

- Virtual machines, containers, and packages
- Debugging and systematic bug hunting
- Profiling and performance
- Data wrangling
- Metaprogramming, automation, and agentic coding

How each week works

Seminar/Workshop (this session, 2 h)

- Concepts, live demos, short activities.
- **Bring your laptop** — you will type along.
- PollEv live questions at pollev.com/fit2109.
- No slides posted in advance — attend to keep up.

Applied class (2 h, separate booking)

- Hands-on lab extending the seminar topic.
- TA-supervised; ask questions freely.
- Several sessions include **assessed demonstrations**.

Typical seminar rhythm

Concept (5–8 min)



Live demo (5–10 min)



Activity / Poll



Assessment Schedule

Four components

Component	Weight	When
In-class demonstrations	15%	Weeks 7, 8, 11, 12
Individual assignments	35%	Weeks 6, 9
In-class quiz/test	15%	Week 9
Group project	35%	Week 14
Total	100%	

Quiz format

The in-class quiz is currently planned as a Moodle quiz. Check Moodle for the final sitting time, platform settings, and permitted resources.

Individual assignments (35% total)

Assignment 1 — due Friday of Week 6

- Shell scripting, editor workflow, Git basics.
- Submitted via **GitLab** (FIT-hosted instance).

Assignment 2 — due Friday of Week 9

- Containers, package management, debugging.

Group project (35%)

Teams

- Team size and formation rules confirmed in the specification.
- Members are expected to come from the **same applied session**.
- Everyone contributes to code, review, testing, and docs.

What you build

A practical tool that brings together shell, Git, containers, testing, automation, secure configuration, and responsible AI use.

Deadlines are strict

Follow the final Moodle instructions for submission, late penalties, extensions, and special consideration.

What to do if you need help

- Check the assessment specification and Moodle announcement first.
- Ask early if submission instructions are unclear.
- Use the official extension and special consideration processes where applicable.

Resources and materials

Where to find everything

Moodle Announcements, assignment specs, grades, quiz.

Ed Discussion All Q&A (join via Moodle link).

FIT GitLab Assessment repositories and project evidence.

Seminar slides Posted on ED.

No required textbook

All material is delivered through seminars and applied classes. Extra reading links are posted on Moodle where relevant.

Setup paths

- Windows: WSL 2 with Ubuntu.
- macOS: Terminal, using Bash for course scripts.
- Linux: Ubuntu recommended; other distributions are fine.
- Optional: Ubuntu Desktop in VirtualBox.

Setup now

Choose one primary setup path *this week*.
Week 1 assumes you can open a terminal and run basic shell commands.

Academic integrity and AI tools

The rule

All submitted work must be your own.
Undisclosed AI use or copying from another student is academic misconduct.

AI tools — use openly

You may use AI assistants **if** you disclose it in your report, can explain every line you submit, and your report reflects on what the AI got wrong.

Why disclosure matters

In this unit we *teach* agentic coding (Week 12). Using AI tools is part of the learning.

Practical test

If a marker asks “walk me through this,” you must be able to do so without the AI tool.
The applied class demonstrations test exactly this.

What we expect from you

Each week

- Attend the seminar with your laptop charged.
- Complete the applied class task — make your hands dirty.
- Run the demos yourself; breaking things is fine.
- Ask questions on Ed, not by email.

Mindset

This unit is hands-on: you will hit errors. Use Google, man pages, and Ed to help you.

What success looks like

A student who passes this unit can work confidently in modern computing environments: using the command line, automation, version control, debugging, containers, and AI-assisted tools to build, diagnose, and deliver software.

The one warning

This unit has more moving parts than most. Students who fall behind in the first three weeks consistently struggle to recover. Start now.

Questions?

If you think of something later — post on **Ed Discussion**.